

Deep Learning

Assignment 4

Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.

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```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report
```

```
# Load dataset
df = pd.read_csv("spam.csv", encoding='latin-1')

# Keep only required columns
df = df[['v1', 'v2']]
df.columns = ['label', 'message']

df.head()
```

	label	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df['label'] = df['label'].map({'ham': 0, 'spam': 1})
```

```
X = df['message']
y = df['label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

```
vectorizer = CountVectorizer(stop_words='english')

X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
```

```
model = MultinomialNB()
model.fit(X_train_vec, y_train)
```

▼ MultinomialNB ⓘ ?

MultinomialNB()

```
y_pred = model.predict(X_test_vec)
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.9827709978463748
```

```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	1202
1	0.96	0.91	0.94	191
accuracy			0.98	1393
macro avg	0.97	0.95	0.96	1393
weighted avg	0.98	0.98	0.98	1393

```
test_messages = [
    "Free entry in a weekly competition to win cash prizes",
    "I'm going home now, will call you later",
    "URGENT! You have won a prize. Call now!",
    "Let's meet tomorrow at college"
]
```

```
test_vec = vectorizer.transform(test_messages)
predictions = model.predict(test_vec)
```

```
for msg, pred in zip(test_messages, predictions):
    print(f"Message: {msg}")
    print("Prediction:", "Spam" if pred == 1 else "Ham")
    print("-" * 60)
```

```
Message: Free entry in a weekly competition to win cash prizes
Prediction: Spam
```

```
-----
Message: I'm going home now, will call you later
Prediction: Ham
```

```
-----
Message: URGENT! You have won a prize. Call now!
Prediction: Spam
```

```
-----
Message: Let's meet tomorrow at college
Prediction: Ham
-----
```